



BORNEO HACKWKND

By: Byte Me :)

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1.0 Introduction

1.1 Background of Case Study

ASLI (ASEAN Stories & Legacy of the Indigenous) is a digital sanctuary dedicated to reclaiming the "original" pulse of Southeast Asia. In the Malay and Indonesian languages, the word *Asli* carries the profound weight of authenticity, referring to that which is original, indigenous, and true to its roots. Currently, the ancestral wisdom of the region's diverse tribes is facing a critical juncture: as globalisation and rapid urbanisation accelerate, the oral histories and traditional customs of ASEAN's communities are being silenced, often trapped in inaccessible academic archives or fading away in isolated villages.

To protect this "Asli" essence, our platform transforms fragmented cultural data into a living, collaborative ecosystem driven by human connection. By bridging the gap between ageing knowledge-keepers and a digital-native younger generation, we provide a space where indigenous legacies are not just stored but celebrated through active storytelling and shared learning. **ASLI** ensures that Southeast Asia's modern future is firmly rooted in its authentic heritage, transforming the fading echoes of the past into a vibrant, accessible heartbeat for future generations.

1.2 Chosen SDG

ASLI is strategically engineered to advance **United Nations Sustainable Development Goal (SDG) 11**, specifically addressing **Target 11.4**: to "strengthen efforts to protect and safeguard the world's cultural and natural heritage." By digitising oral histories and indigenous wisdom, our platform serves as a modern guardian for Southeast Asia's disappearing legacies. Furthermore, the project directly contributes to **SDG 4 (Quality Education)** by transforming fragmented cultural data into accessible, inclusive, and high-quality educational resources, ensuring that ASEAN's rich heritage remains a foundational pillar for lifelong learning.

1.3 Problem Statement

The "original" heartbeat of Southeast Asia is at a critical tipping point. Despite the presence of over 4,000 indigenous languages and traditions in ASEAN, this cultural wealth is rapidly disappearing due to a significant "documentation gap" and a generational disconnect.

The Challenges:

- **Academic Silos:** Existing cultural knowledge is often "trapped" in dense academic research or restricted to isolated local communities, making it inaccessible to the public, which hinders efforts to preserve and share these valuable traditions with a broader audience.
- **The "Digital Invisibility" of Indigenous Voices:** Current digital platforms prioritise globalised content. Indigenous communities lack the **user-friendly, mobile-first tools** required to record and own their own narratives, which hinders their ability to share their stories and cultural heritage effectively in the digital space.
- **Intergenerational Disconnect:** While **97% of ASEAN youth** are digital natives who prefer interactive learning, there is a severe lack of engaging platforms dedicated to regional heritage, which prevents these youth from accessing and learning about their cultural history effectively.

Without a modern bridge to connect the past to the present, Southeast Asia's authentic (Asli) indigenous legacy risks fading into oblivion.

1.4 User Persona

1. The Digital Learner (Knowledge Seeker)

Attribute	Profile
Name	Adi, 21
Role	University Student (History/Anthropology)
Location	Kuala Lumpur, Malaysia
Goals	To find authentic primary sources for his thesis on Sarawakian heritage and reconnect with his roots.
Pain Points	Most online info is generic or tourist-focused; he can't travel to remote longhouses easily to interview elders.
App Usage	Uses the Map to find specific tribe locations and the Folktales section to listen to authentic narrations.

2. The Cultural Guardian (Knowledge Keeper)

Attribute	Profile
Name	Indai Rani, 68
Role	Skilled Weaver and Oral Historian
Location	Kapit, Sarawak (Iban Longhouse)
Goals	Her goal is to ensure that the younger generation remembers the significance of Pua Kumbu weaving and Iban rituals.
Pain Points	The youth in her village are moving to cities; she worries her knowledge will die with her.
App Usage	She uses the Stories feature to record her voice and uploads photos of her traditional arts, such as weaving, for preservation.

1.5 Objectives

The primary goal of ASLI (**ASEAN Stories & Legacy of the Indigenous**) is to revitalise the region's heritage through a living digital ecosystem.

Our specific objectives are:

- I. **Centralise Cultural Assets:** To build a unified digital sanctuary that preserves and organises ASEAN's diverse indigenous stories, oral histories, and traditional knowledge.
- II. **Empower Community Voice:** To provide a **bi-directional storytelling platform** where indigenous members and cultural practitioners can share and verify their traditions, moving away from "outsider-only" documentation.
- III. **Modernise Cultural Education:** To create an **interactive learning** experience utilising map-based exploration and multimedia that resonates with the consumption habits of younger generations.
- IV. **Foster Regional Solidarity:** To promote cross-cultural understanding by highlighting the shared roots and unique differences across the ten ASEAN nations, including their diverse languages, traditions, and historical experiences that shape their identities.
- V. **Demonstrate Technical Feasibility:** To design and deploy a **high-fidelity interactive prototype** that proves how role-based interaction (learner, storyteller, verifier) can create a sustainable cycle of cultural preservation.

2.0 Review of Existing Solutions

1. Google Arts & Culture

Google Arts & Culture is a powerhouse for digital preservation, hosting over **6 million high-resolution images**. However, its structure is rooted in "Digital Cultural Colonialism".

- **Artifact-Centric:** A 2021 study in *Digital Scholarship in the Humanities* found that the platform heavily prioritises major urban institutions and Western-centric collections.
- **Representation Gap:** While it has partnered with the ASEAN Centre for Biodiversity for natural heritage, the representation of the region's **400+ diverse ethnic groups** remains largely limited to national masterpieces rather than daily indigenous life, which diminishes the visibility of their unique cultural practices and narratives.
- **Static Experience:** It functions as a digital museum where users are "visitors" rather than "contributors", preventing the communal storytelling vital to ASEAN cultures.

2. UNESCO Intangible Cultural Heritage (ICH) Platform

UNESCO provides the global gold standard for documentation, but it operates within a rigid, bureaucratic framework.

- **The "Inscription" Barrier:** To be recognised, a tradition must go through a complex state-level nomination process. This process often excludes smaller, marginalised tribes who lack the political capital to be "inscribed".
- **Academic vs. Accessible:** Most data are presented in formal reports and policy documents. UNESCO's own research indicates that **nearly 50% of the world's scripts**, many from minority ASEAN languages, are not supported by major digital platforms, making their data unreadable to the very communities that own it.
- **Passive Learning:** While they have "Dive into ICH" interfaces, the connection between elements is thematic and academic, not narrative-driven or interactive for a younger, mobile-first audience, which limits engagement and understanding of the content among this demographic, particularly in how it relates to their cultural heritage and interests.

3. Cultural Tourism Websites

Sites like "Wonderful Indonesia" or "Tourism Malaysia" are highly effective at visual marketing but lack preservation depth, which can lead to the oversimplification of complex cultural narratives and the potential erasure of the communities' authentic voices.

- **Commodification Risk:** A 2025 report in *Modern Diplomacy* warns that tourism-driven digitalisation often reduces sacred ancestral stories to "mere visual decoration" or commercial content without community consent.
- **Surface-Level Representation:** Minority tribes (like the *Orang Asli* in Malaysia or the *Dayak* in Borneo) are often featured as "exotic attractions" rather than knowledge-holders, leading to a loss of cultural context.
- **Zero Interaction:** There are no tools for intergenerational knowledge transfer; once the tourist leaves the site, the "engagement" ends, resulting in a missed opportunity for cultural exchange and the preservation of traditional knowledge within the community.

2.1 Gap Analysis: Why ASLI is Necessary

The table below summarizes the critical gaps in the current landscape:

Feature	Google Arts & Culture	UNESCO ICH	Tourism Sites	ASLI (Our Solution)
Interactive Tribe-Specific Learning	✗	✗	✗	✓
Community-Led Storytelling	✗	✗	✗	✓
Map-Based Exploration System	⚠ (Limited)	✗	✗	✓
Crowdsourced Documentation	✗	✗	✗	✓
Indigenous Language Support	✗	⚠ (Limited)	✗	✓

3.0 Methodology

Our team adopted a **Human-Centred Design (HCD)** approach, utilising the five-stage **Design Thinking** framework to ensure ASLI is both culturally sensitive and technologically intuitive.

Stage 1: Empathize (User Research & Ecosystem Mapping)

We conducted a stakeholder analysis to identify the heartbeat of the platform. Our research revealed a significant "Engagement Gap": while **85% of ASEAN youth** use digital platforms for daily learning, only a fraction find indigenous content that feels relevant or modern.

- **Key User Personas:**
 - **The Digital Learner:** Students seeking immersive, non-academic cultural content.
 - **The Heritage Guardian:** Indigenous elders and community members with oral histories to share.
 - **The Educator/Researcher:** Professionals needing centralised, verified primary sources.
 - **The Cultural Enthusiast:** Travellers or hobbyists looking for authentic (Asli, meaning original or genuine) regional depth.

Stage 2: Define (The Core Challenge)

We distilled our research into a single problem statement: *How might we transform "static" cultural data into a "living" collaborative ecosystem that appeals to a mobile-first generation?* We identified that for a solution to be successful, it must be **Hyper-Local** (tribe-specific), **Bi-Directional** (users can both consume and create), and **Gamified** (to sustain engagement).

Stage 3: Ideate (Feature Architecture)

To address the defined problem, we conceptualized a multi-layered interface that prioritizes **Visual Discovery** over text-heavy searching.

- **Interactive Cartography:** A "Global-to-Local" map navigation system.
- **Role-Based Interaction:**
 - **Learner:** Consumes knowledge through "Story Cards" and interactive quizzes.
 - **Storyteller:** Contributes oral histories via voice notes or short-form video.
 - **Contributor:** Uploads digital twins of physical artefacts or traditional crafts.
 - **Verifier:** A community-led moderation layer to ensure historical and cultural accuracy.

Stage 4: Prototype (Figma UI/UX Development)

We developed a high-fidelity prototype in **Figma**, focusing on a "Vibrant Heritage" aesthetic that uses earthy tones paired with modern UI elements. Key screens include:

- **The ASEAN Gateway:** An interactive map for macro-selection.
- **The Tribal Matrix:** Precision map markers indicating indigenous territories.
- **The Wisdom Hub:** A role-selection dashboard tailored to the user's intent.
- **The Living Library:** A multimedia repository for stories, music, and imagery.

Stage 5: User Flow & Logic

We engineered a seamless, friction-free path from discovery to contribution.

Standard Journey:

Launch - Select Country (e.g., Vietnam) - **Identify Tribe** (e.g., Hmong) - **Choose Role** (Learner) - **Engage** (Watch Video/Read Story)

4.0 Results

The project successfully transitioned from a conceptual framework to a **high-fidelity functional UI prototype**. By applying design thinking and leveraging AI-assisted tools, we have met all core objectives outlined at the start of the development phase.

4.1 Achievement of Objectives

The following table summarizes the status of our project goals:

Objective	Outcome	Status
Interactive Map System	Designed a GIS-inspired interface that allows users to navigate cultural heritage by geography rather than just text.	✓
Multi-Role Participation	Implemented three distinct user roles (Learner, Storyteller, and Verifier) to create a self-sustaining ecosystem.	✓
Digital Storytelling	Developed a dedicated "Folktales & Stories" module featuring audio-visual integration for oral history preservation.	✓
Integrity & Verification	Built a reference-based submission system to ensure that uploaded cultural data is authentic and credible.	✓

4.2 Summary of Outcome

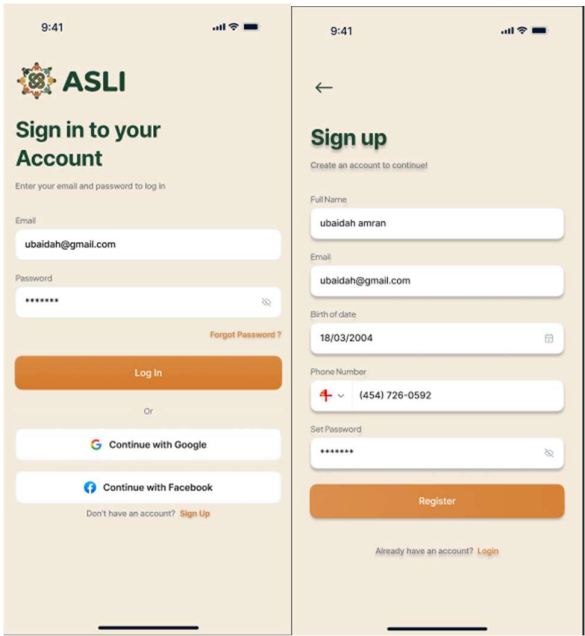
The primary outcome of this work is a **validated digital blueprint** for regional cultural preservation.

- **Technical Success:** We successfully integrated complex features (mapping, multi-modal storytelling, and verification) into a cohesive, user-friendly mobile interface.
- **Impact Potential:** The prototype demonstrates that digital technology can act as a "living archive," making indigenous knowledge accessible to the youth while giving elders a platform to share their wisdom.
- **Proof of Concept:** While currently in the prototype stage, the seamless flow between modules proves that a community-driven cultural hub is both technically feasible and socially necessary for the ASEAN region.

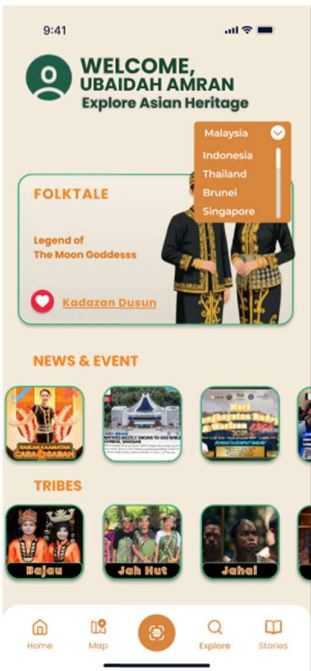
4.3 Future Validation

While the design objectives were met, the next phase will involve **Usability Testing (UT)** with members of the Iban community and regional students. This will provide qualitative feedback to further refine the "Verification" algorithm and optimize the "Story-Sharing" UX for older users.

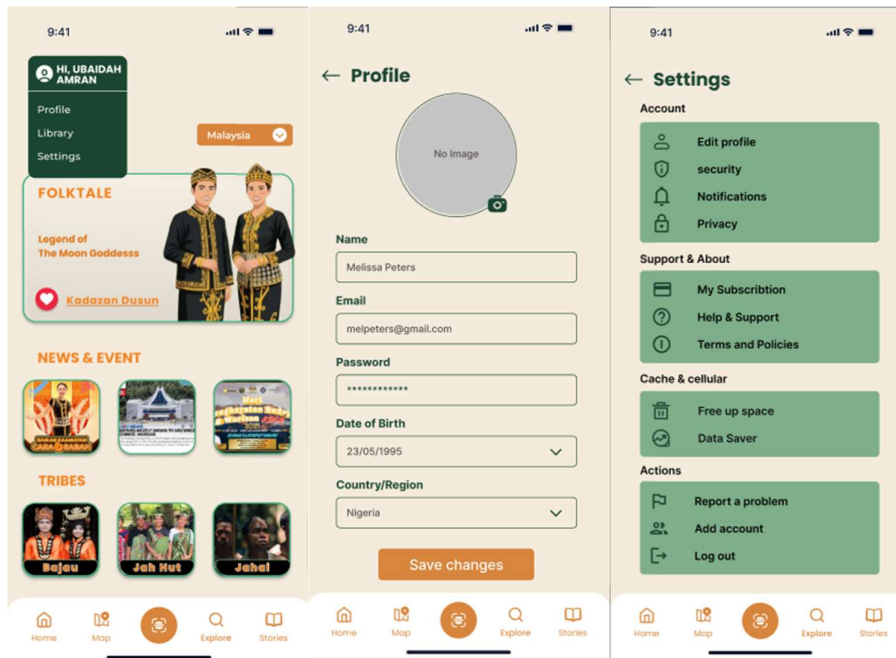
5.0 Prototype Showcase



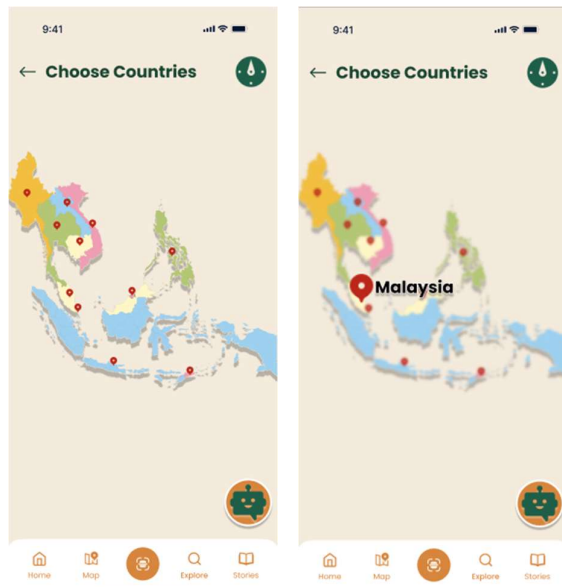
Sign in and Sign up page: For security purposes



Home Page
This page shows the h



From the Home Page, it has a button on the user's profile picture and has 3 options, which are: Profile, Library and Settings. For the profile page, user can edit their profile and save it. Next, in the settings page, the user can see the options like the picture above.



The user can choose which country has a tribal community and can choose which tribe they want to learn about.



If the user does not know the tribe that they want to search, but just has a certain type of that tribe, they can ask the AI chatbox, and the AI will give the answer based on the criteria that the user give.

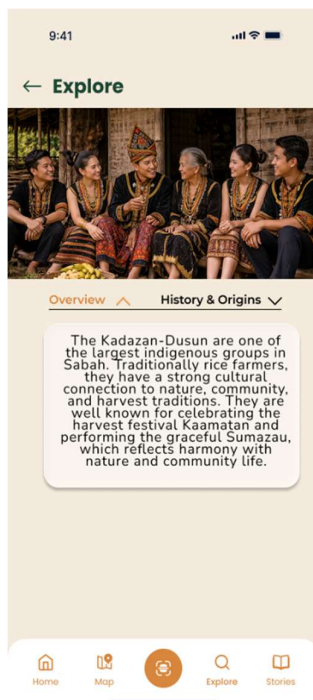


After the user chooses the tribe that they want to learn or search, the user can choose 3 options, which are: Explore, Share and Stories.



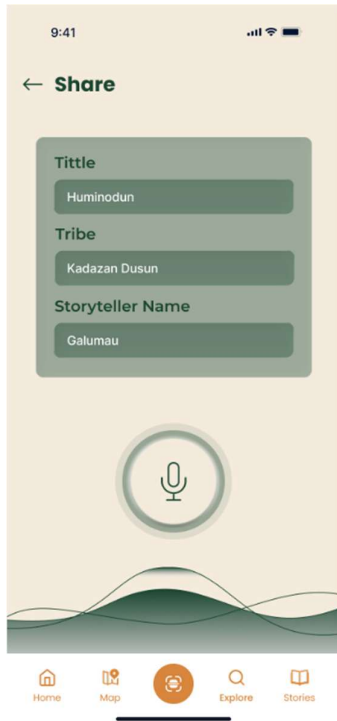
ASLI Lense (Scan Page)

An AR or image-recognition tool that lets users scan cultural artifacts (like the Mah Meri mask). Once scanned, the app provides a pop-up explanation of the artifact's cultural significance, origins, and spiritual meaning.



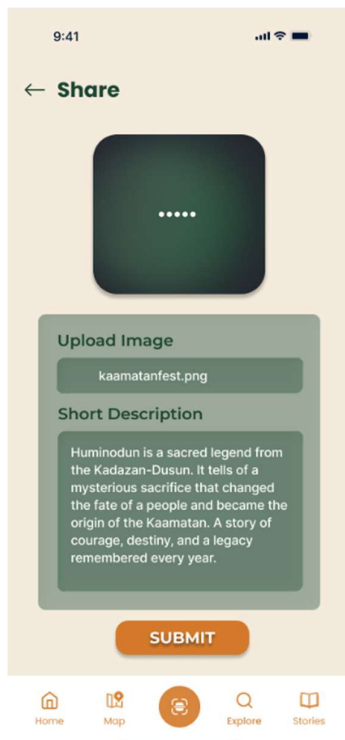
Tribe Profile (Explore Pages)

An educational discovery screen dedicated to a specific indigenous group. It features a high-quality header image of the community and uses a tabbed system so users can switch between a quick "Overview" of their culture and a deeper dive into their "History & Origins."



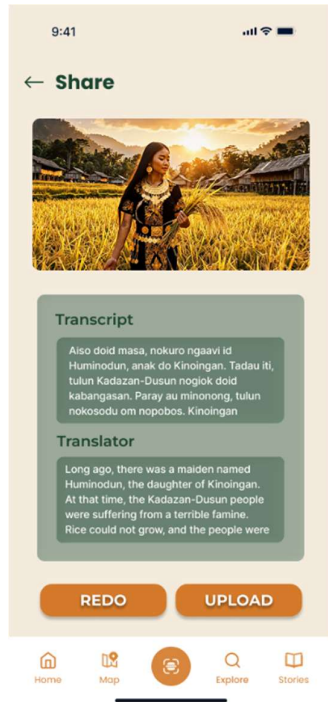
Record Audio (Share Page)

The starting point for contributing a new oral history. Users input the metadata like Title, Tribe, and Storyteller Name before using the prominent microphone button to record the story directly into the app.



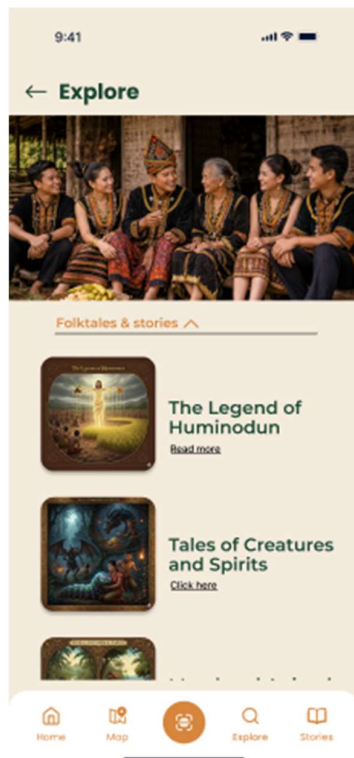
Submission (Submit Page)

The step in the upload flow. Users can attach a relevant cover image (like kaamatanfest.png) and write a short description or summary of the legend before hitting "Submit" to publish it to the database.



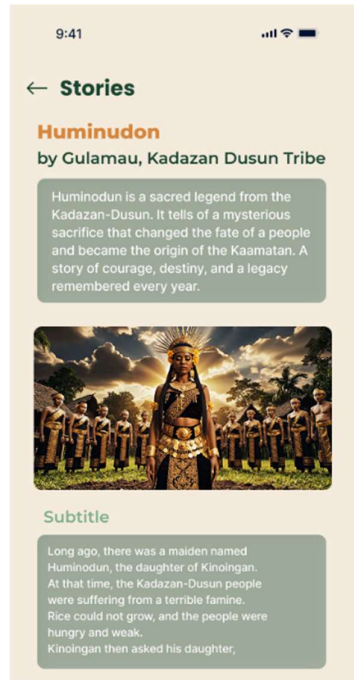
Transcript & Translation Review (Upload Page)

A review screen that acts as an AI-powered translation bridge. It displays the audio's original transcribed text (e.g., Kadazan-Dusun) alongside the translated English version, allowing the user to verify accuracy before uploading.



Published Story View (Variety of Stories Page)

This interface make help user to choice what type of story they are interested. User can “click more” to read or “click here” to see how many story under each category. Then enjoy it.



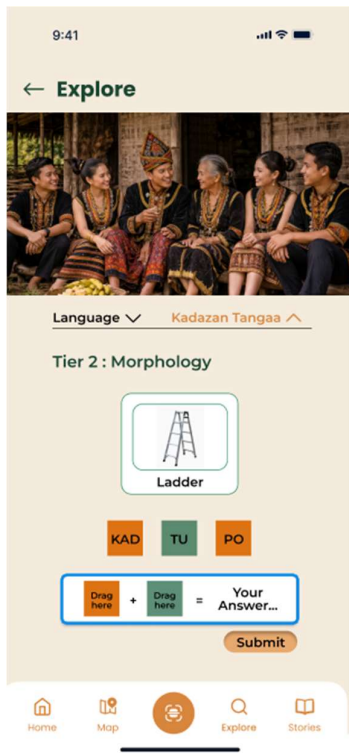
Published Story View (Stories Page)

The reading interface for users consuming the content. It displays the story title, storyteller credit, a large feature illustration, and clear, readable blocks of translated text for an immersive storytelling experience.



Type of Language for Indigenous (Language Page)

From here users can explore any type of languages for indigenous and learn it with impactful methods. Such as phonology, morphology, syntax, context-based learning and garden or memory (revision).



Learning Method (Language Page)

This show user learning in tier 2 which is morphology technique. User need to identify which combination block of syllables is correct base on the picture. Block of syllables is provided but have tricky syllables not use for that word. So, user need to choice carefully.

6.0 AI Acknowledgement

We acknowledge the use of advanced digital tools and Artificial Intelligence to enhance the creative and structural development of **ASEAN CultureBridge**. These tools were instrumental in bridging the gap between our conceptual vision and the high-fidelity prototype, allowing us to effectively visualise and iterate on our ideas before final implementation.

Disclosure:

- **Gemini**
- **ChatGPT**
- **ElevenLabs**
- **Figma**
- **Grammarly**

Usage:

Tool	Category	Specific Usage & Contribution
Gemini	Generative AI & Strategy	Utilized for conceptual strategy , brainstorming the project framework, and generating high-fidelity visual assets . It also assisted in drafting and structuring the technical report.
ChatGPT	Conversational AI & Research	Utilised for UX writing , refining microcopy (buttons, labels), and synthesising user research data to build persona profiles.
ElevenLabs	AI Audio Synthesis	Employed for generating realistic narration and voiceovers for the oral history sections, providing a high-quality auditory experience for the prototype.
Figma	Design & Prototyping	The primary platform for building the User Interface (UI) , mapping out application flows, and creating the interactive high-fidelity prototype.
Grammarly	Linguistic Optimization	Used for final editorial oversight , ensuring that all project documentation and report content were professional, clear, and grammatically precise.

7.0 Conclusion

ASLI is more than a digital archive; it is a movement to reclaim the narrative of Southeast Asia's indigenous peoples. By integrating **design thinking** with geographic information systems (**GIS**) and **community-led storytelling**, we have built a bridge that allows ancient wisdom to cross over into the digital age. Our prototype demonstrates that when technology is used to empower rather than commodify, it can revitalise disappearing languages and traditions for a new generation.

Moving forward, we envision **ASLI** evolving into a sentient cultural ecosystem. By incorporating **AI-driven translation** to break language barriers and **Recommendation Engines** to personalise cultural discovery, we aim to transform ASEAN's fragmented heritage into a unified, thriving digital hub. By doing so, we not only remember but also embody the "original" (asli) spirit of the region.